Istituto Nazionale di Fisica Nucleare
Laboratori Nazionali di Frascati

Roma, 19 novembre 2024

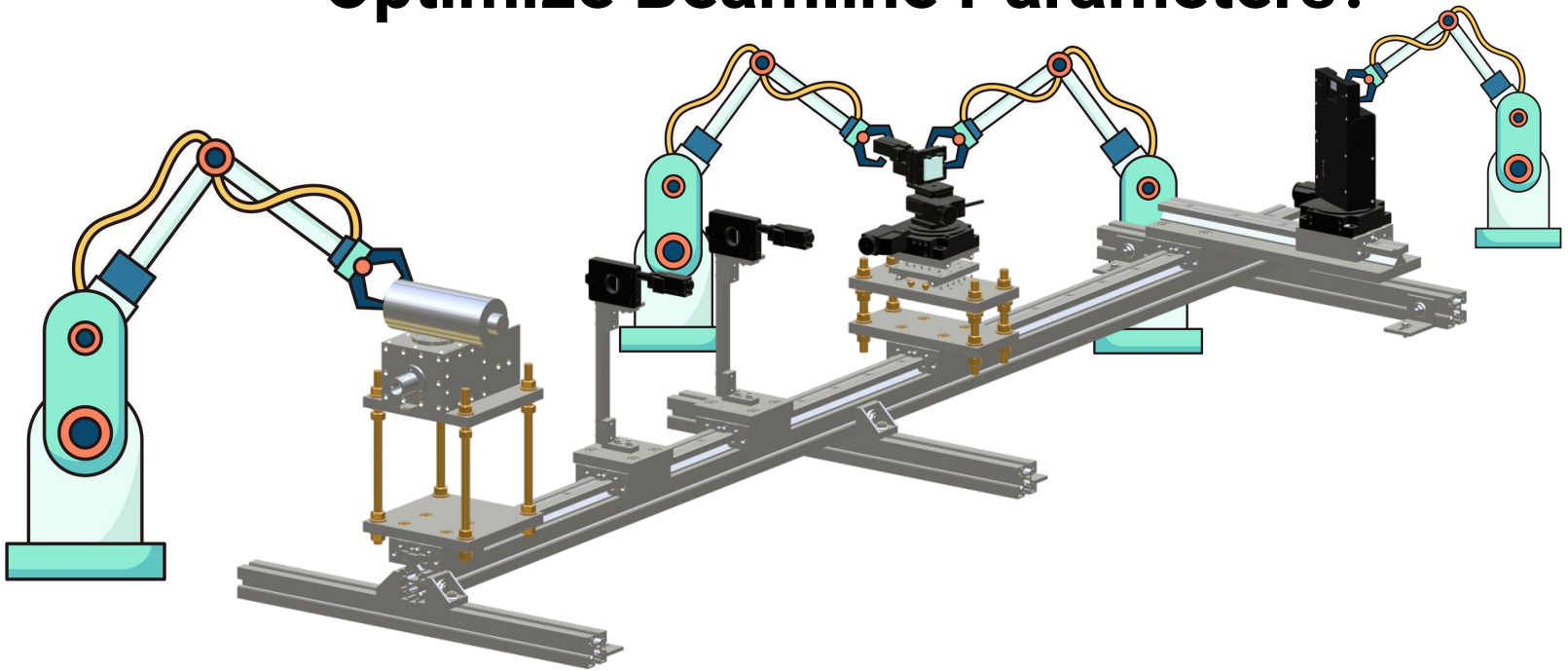
PROPOSAL GRANT GIOVANI CSN5 2024

OPTIMAL

automated Optimization of x-ray beamline Parameters using experimenT-Informed MAchine Learning methods

Simone Manti, LNF INFN

How can Machine Learning methods be used to Optimize Beamline Parameters?

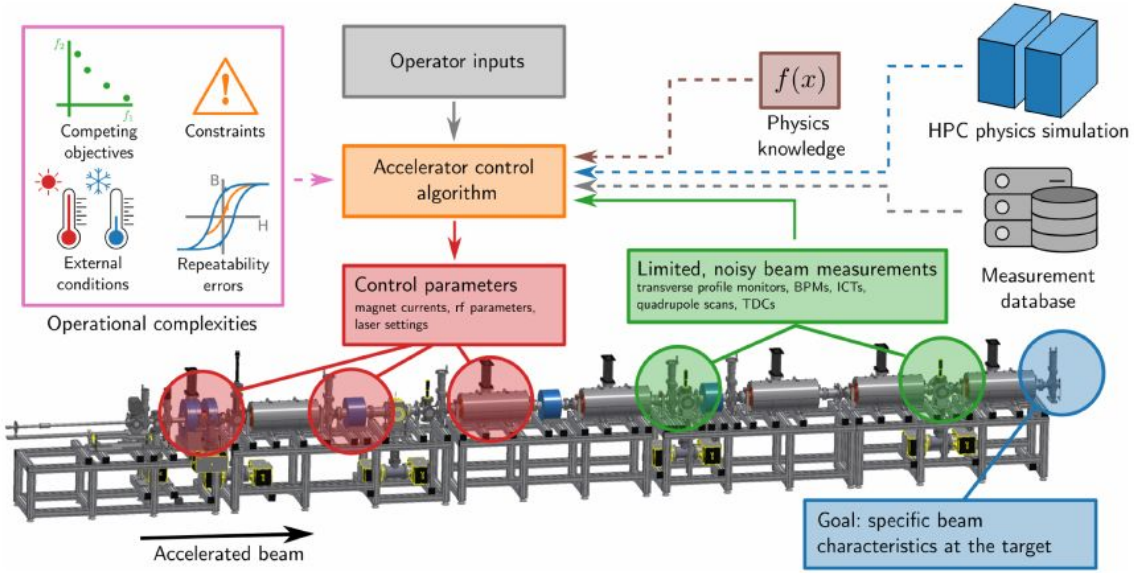


Why is important to develop an Automated Optimization of beamline parameters?

- ⏳ Minimize **Time** from setup to data collection
- 💡 Reduce **Operational Costs**
- 📈 Enhance **Data Quality** & **Reproducibility**
- 🧪 Tailor to Specific **Research Needs**
- ⚙️ Limit **Human Errors**

Increased Use of ML methods for Experimental Optimization of beamlines and accelerators

- **Particle Accelerators**: Optimize operational parameters
- **Free-Electron Lasers**: Refine laser pulse properties
- **Plasma Accelerators**: Adjust beam dimensions



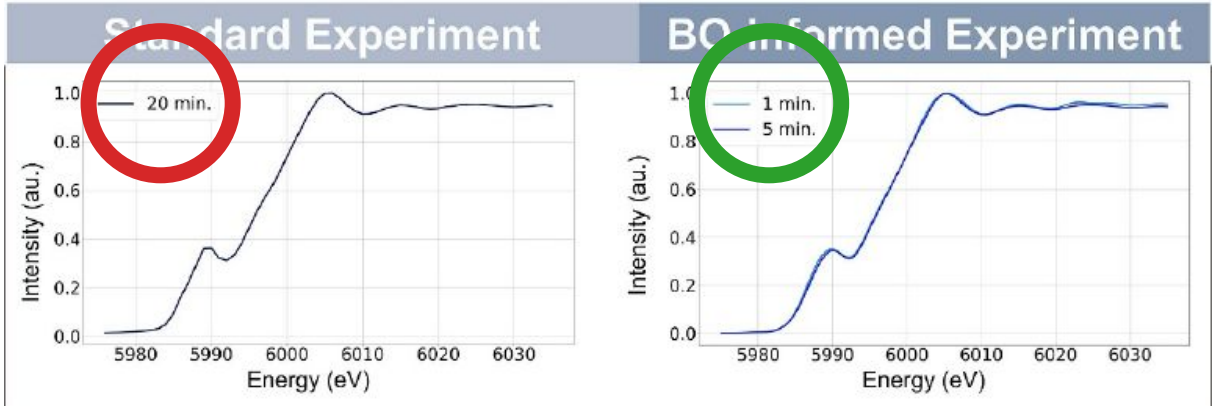
Roussel, Ryan, et al. "**Bayesian optimization algorithms for accelerator physics**." *Physical Review Accelerators and Beams* 27.8 (2024)

How?

- **Bayesian Optimization (BO)**
- **Reinforcement Learning (RL)**

Status in X-ray Beamline Experiments

- **Diverse Techniques:** Each one (e.g. XAS, XES,...) with unique requirements
- **Setup demands:** Different constraints for **Synchrotron** vs. **Tabletop Setups**
- **Limited use of ML optimization:** it requires specialized models and expertise

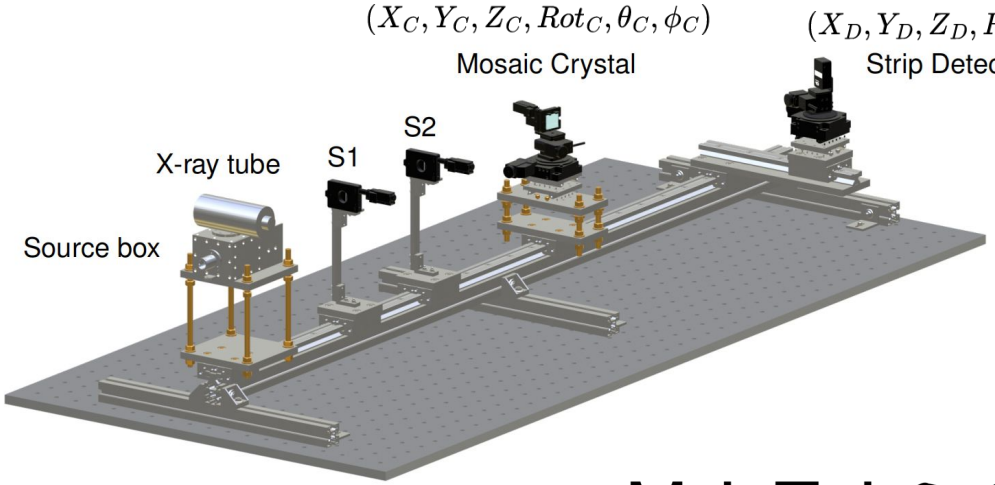


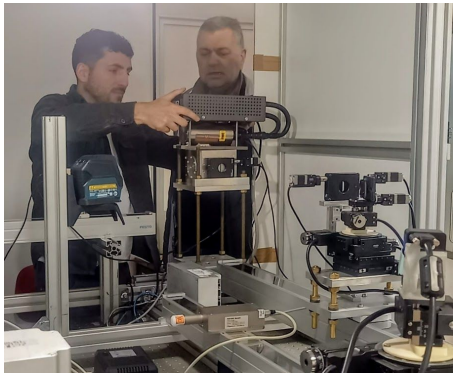
e.g. Bayesian Optimization independently to reduce XANES measurement time by selecting optimal data points.

Cakir, Cafer Tufan, et al. Machine Learning: Science and Technology 5.2 (2024): 025037.

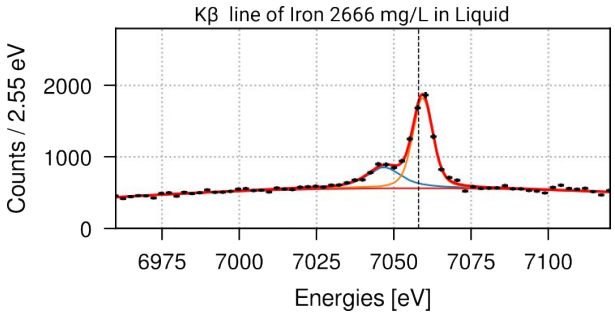
Integrating ML-driven techniques within an Optimization Framework to enhance efficiency in X-ray experiments is the core concept behind OPTIMAL ✨

Challenges for XES operations: VOXES at LNF as a case study





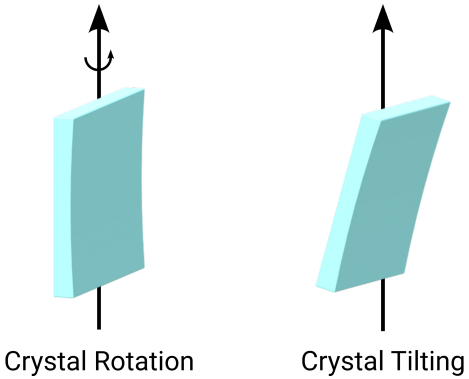
MITIQO

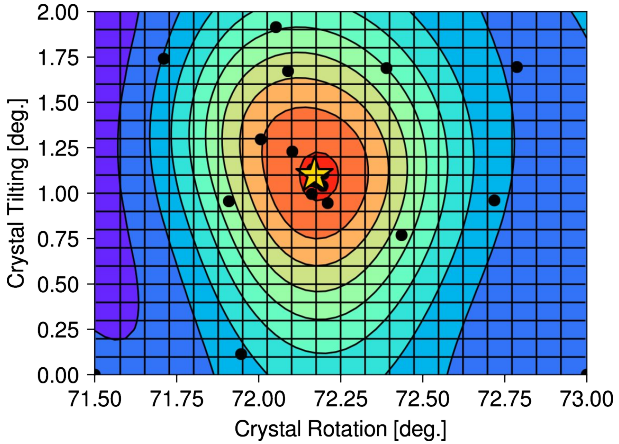


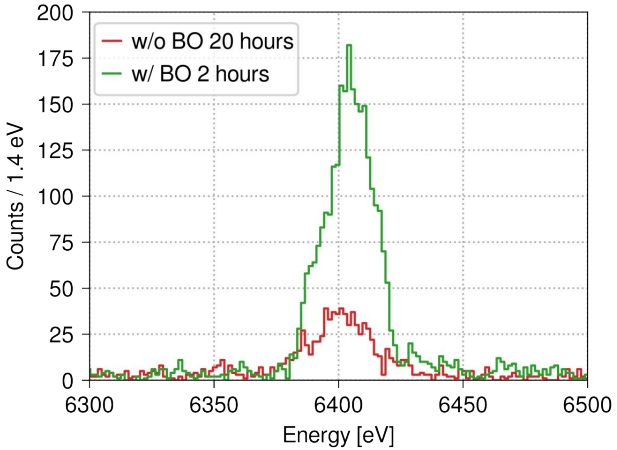
- **Typical Operations:**
  - Move to align $\mathbf{x_C}$ and $\mathbf{x_D}$
  - **Crystal** fine alignment
  - **Detector** fine alignement
  - Set $\mathbf{S_1}$ and $\mathbf{S_2}$ for **Rate** and **Resolution**
- Rate ~Hz, challenging for **High-throughput** measurements and **Low Concentrations**!
- Performed optimization for **Background Reduction**

Manti, Simone, *et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." Condensed Matter 9.1 (2024): 19.*

Bayesian Optimization Reduces Operation Time

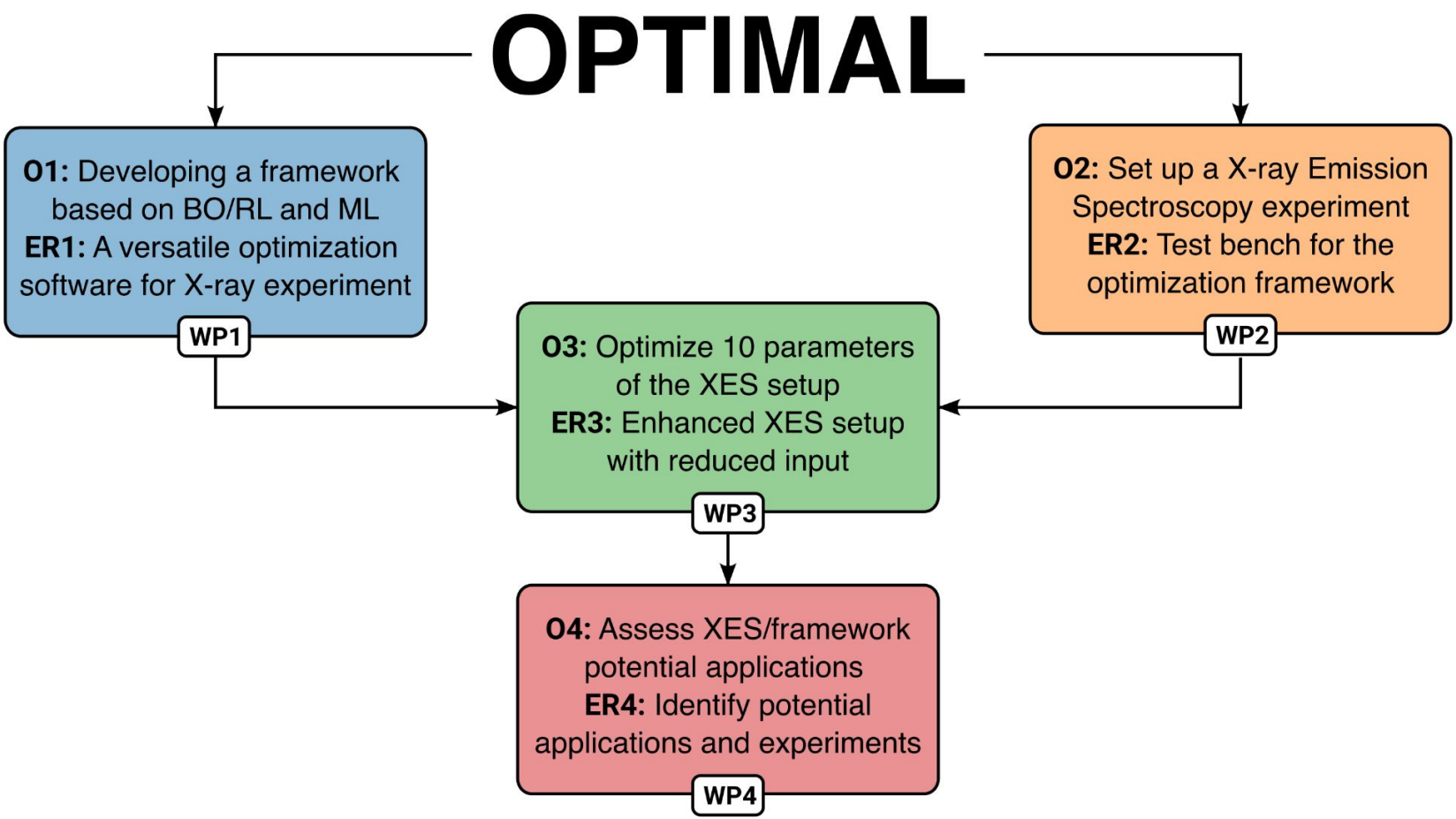




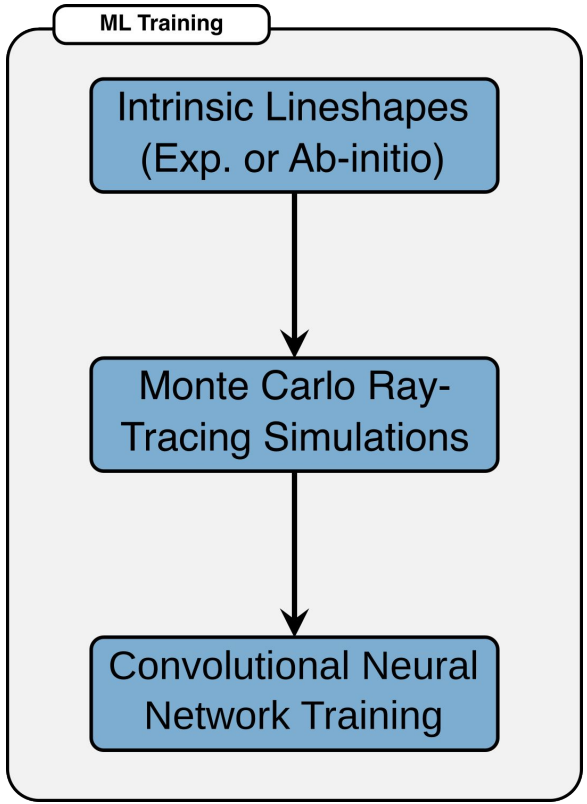


- **Improved Alignment**: Feasibility study optimizing crystal **Rotation** and **Tilting** for the **Iron Kα** emission line, achieving precise alignment.
- **Faster Convergence**: Bayesian Optimization achieves convergence in just **20 iterations**, surpassing manual operations.
- **Enhanced Rate and Efficiency**: Optimized parameters result in higher rate intensity and reduced operation time from **20 hours** to under **2 hours**.

OPTIMAL


O1: Developing a framework based on BO/RL and ML
ER1: A versatile optimization software for X-ray experiment
WP1
O2: Set up a X-ray Emission Spectroscopy experiment
ER2: Test bench for the optimization framework
WP2
O3: Optimize 10 parameters of the XES setup
ER3: Enhanced XES setup with reduced input
WP3
O4: Assess XES/framework potential applications
ER4: Identify potential applications and experiments
WP4

Framework based on ML model trained on the Ray-Tracing Simulations of the Experiment (O1,WP1)



Simulations of Intrinsic Emission Line Shapes:

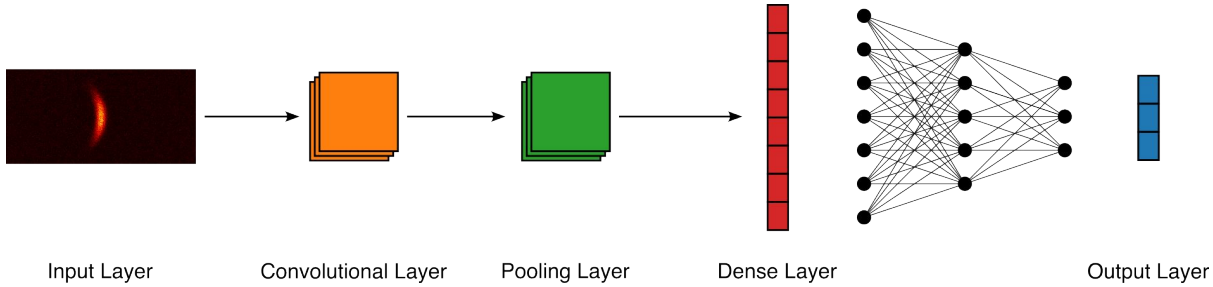
- Derived from **Experimental Linewidth** data
- Based on **Ab Initio** calculations

Ray-Tracing Simulations of the XES Setup:

- Utilizing **SHADOW-XOP** with the **OASYS** environment

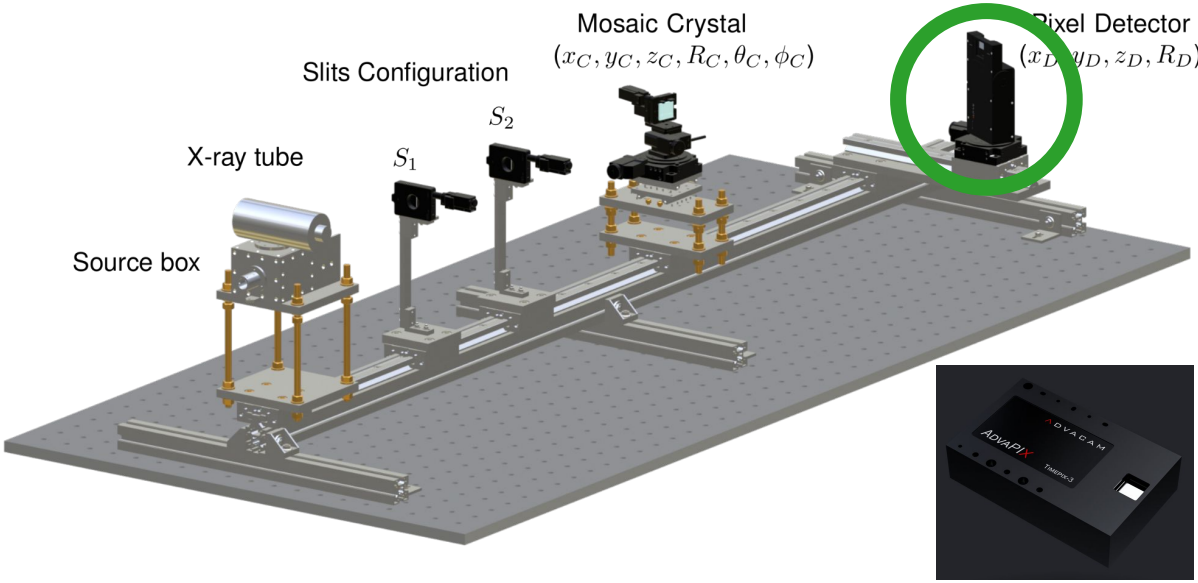
Training a Convolutional Neural Network:

- For **Features Extraction** as input for the **Optimization loop**

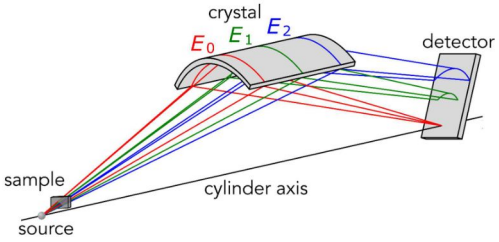


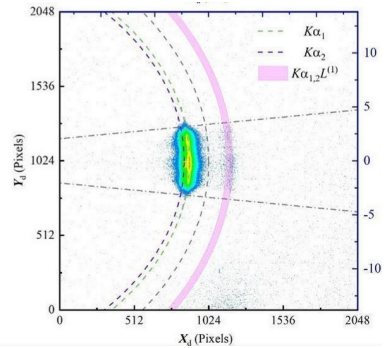
Inclusion of a Pixel Detector in the XES setup (O2,WP2)

- Exploit **ML capabilities** on **2d Image** compared to a **1d Histogram**
- Improve spectrum acquisition for alignment in the **Vertical Direction**



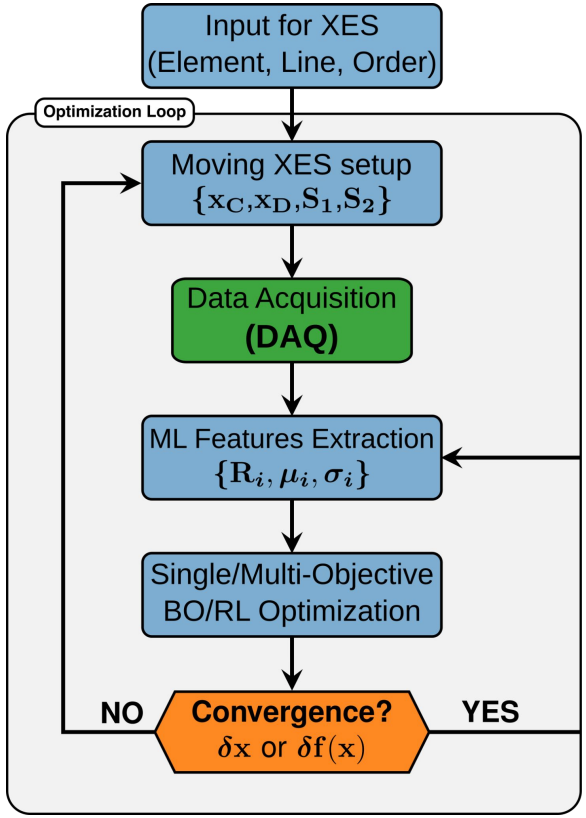
AdvaPIX Timepix3 from ADVACAM)



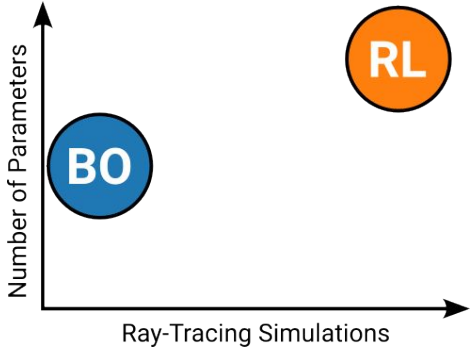


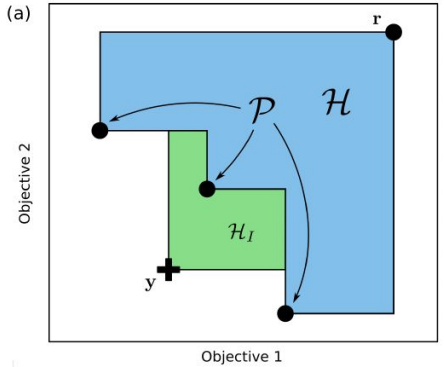
Wang, W., et al. *Spectrochimica Acta Part B: Atomic Spectroscopy* 215 (2024): 106918.

Optimization Loop to iteratively find the best Parameters (O3,WP3)

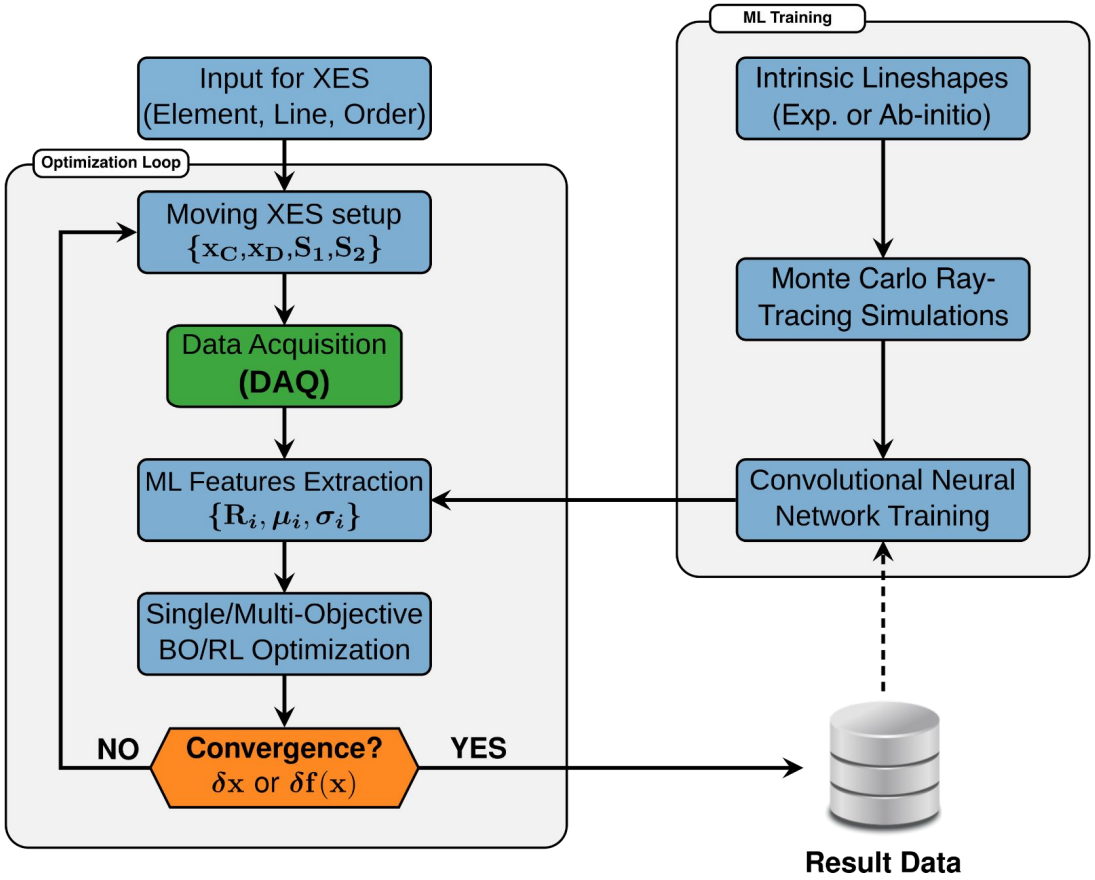


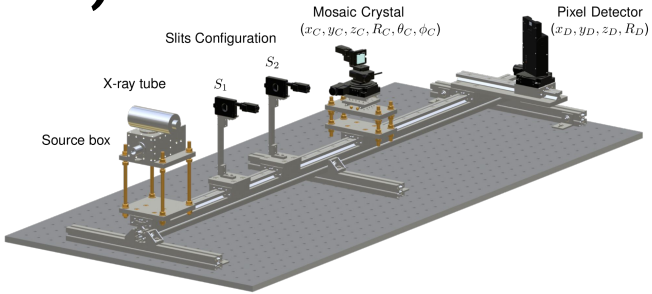
- Controlling Setup and Data Acquisition (DAQ)
- **Features Extraction** from Trained ML Models
- Complementary and Interoperable **Optimization Methods**:
  - **Bayesian Optimization**
  - **Reinforcement Learning**
- **Optimization Goals:**
  - **Single-Objective**
  - **Multi-Objective**





Optimization Workflow of OPTIMAL (O3,WP3)





- Saving data in a **Database** to eventually training on **Historical Data.**
- Optimize **10 Key Parameters** of the XES Setup.
- Enhanced **Adaptability** to specific use cases, for a broader range of **Applications**.

Applications and Synergies (O4,WP4)

INFN Scientific Commissions and Collaborations:

- **VOXES->OPTIMAL** (LNF)
- **DAΦNE-Light**, **EuAPS** (LNF), **SPHINX** (CSN5)
- **EuPRAXIA** (LNF)

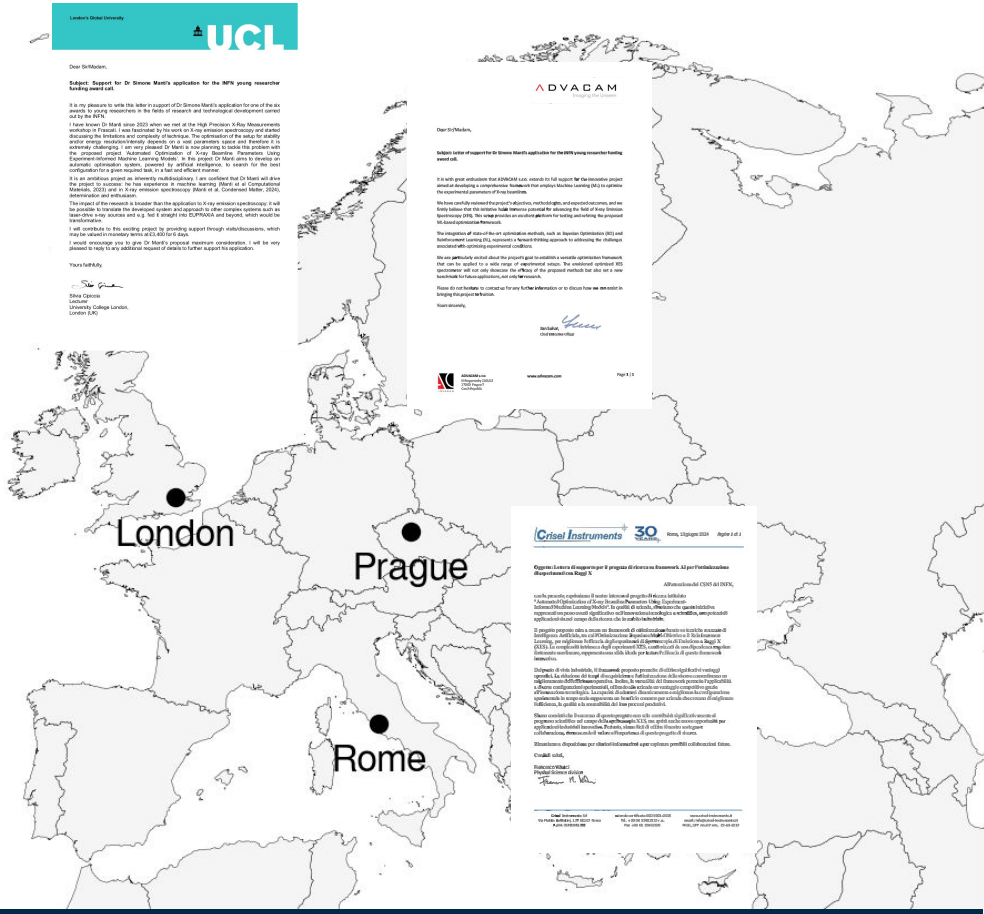
National and International Research Institutions:

- **Advanced X-Ray Imaging Group** (UCL)

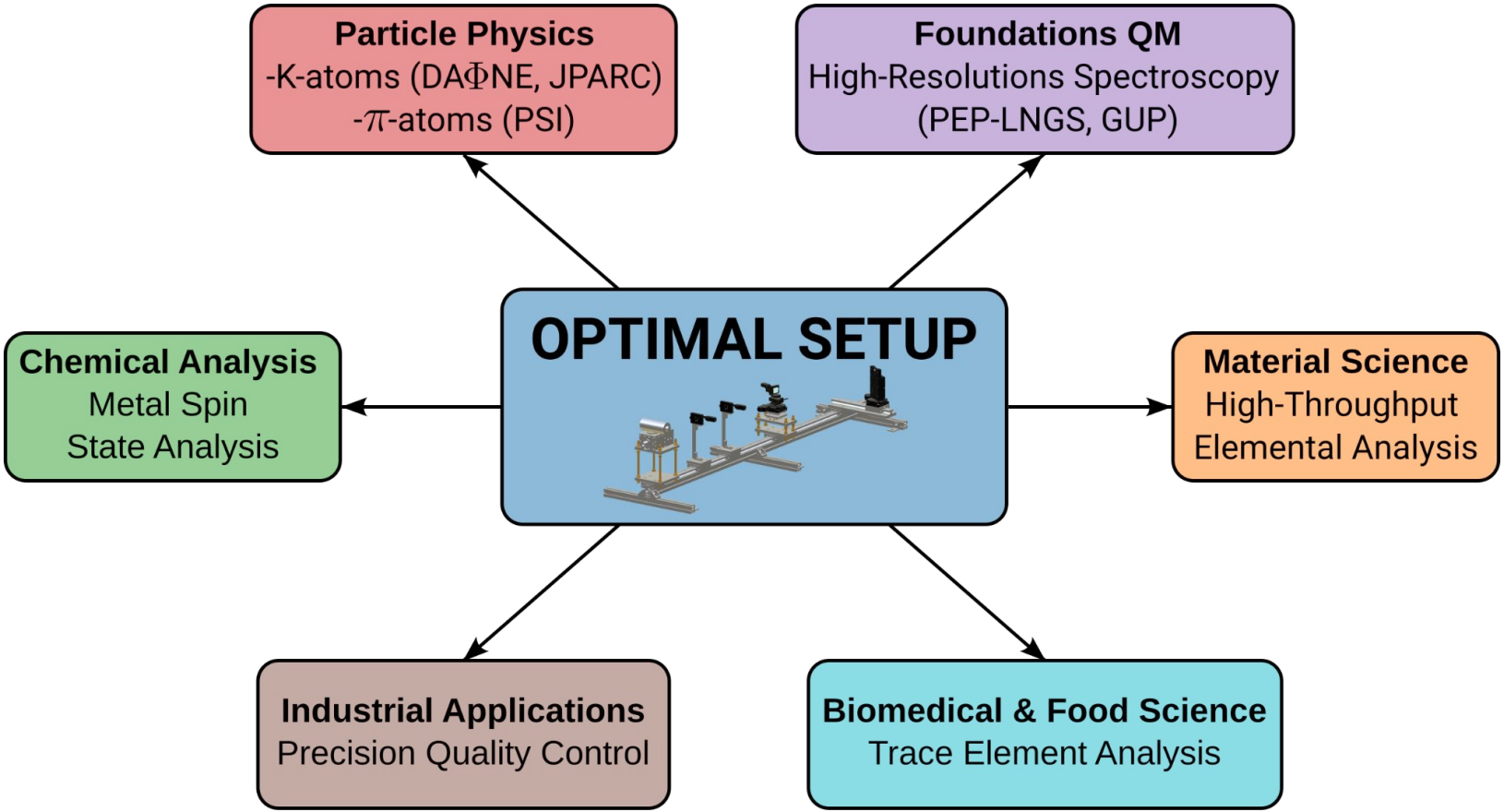
Industry and Private Sector:

- **ADVACAM** (Prague)
- **CRISEL** (Rome)

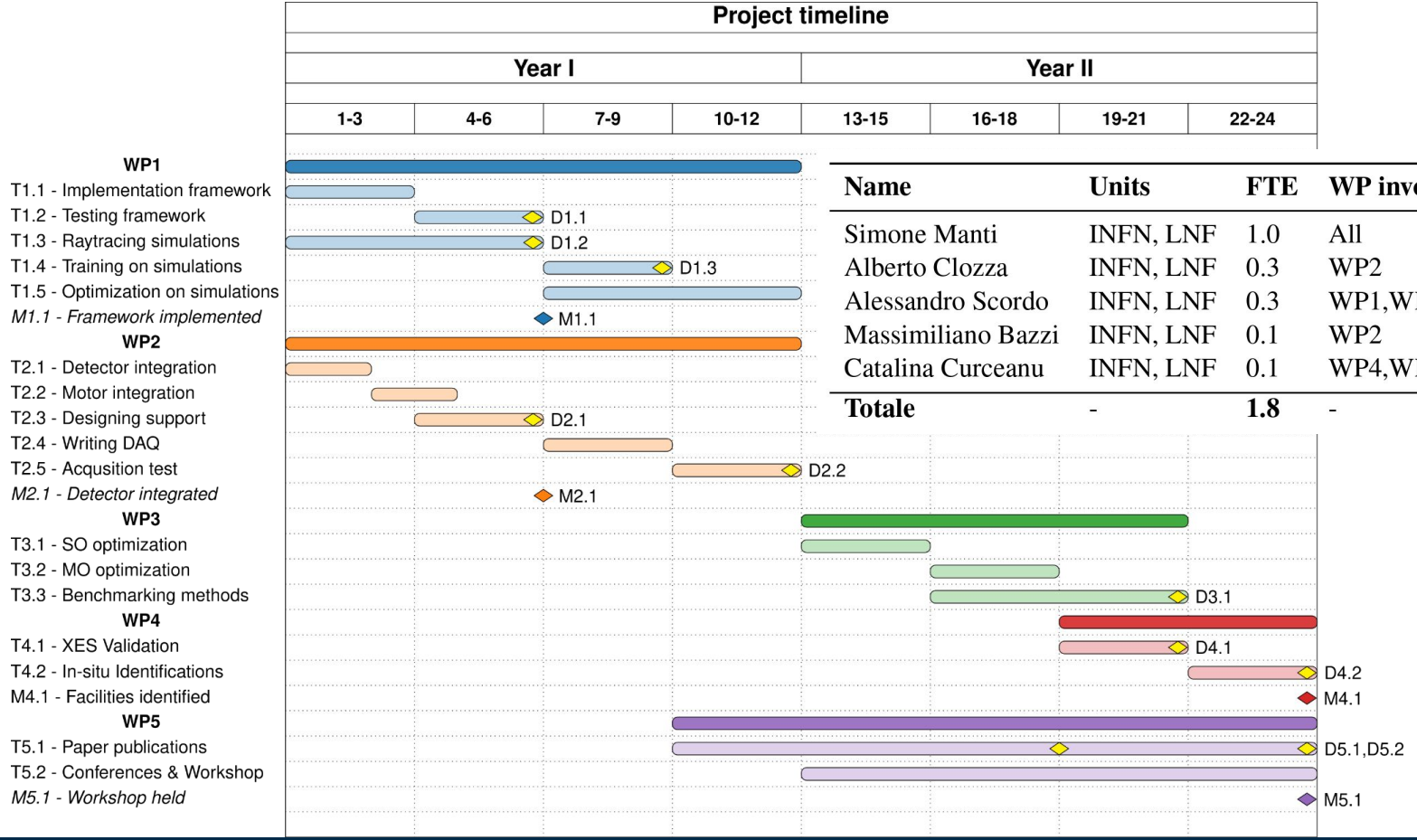
→ **Digital Twins**



Impact of the OPTIMAL Setup



A Work Package for every Objectives plus Dissemination

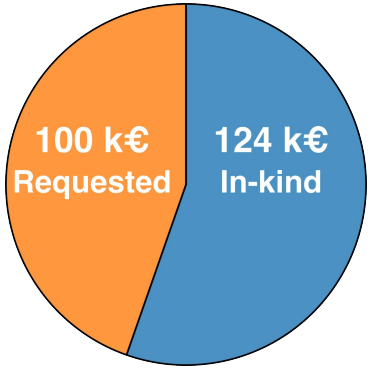


Name	Units	FTE	WP involved
Simone Manti	INFN, LNF	1.0	All
Alberto Clozza	INFN, LNF	0.3	WP2
Alessandro Scordo	INFN, LNF	0.3	WP1,WP3
Massimiliano Bazzi	INFN, LNF	0.1	WP2
Catalina Curceanu	INFN, LNF	0.1	WP4,WP5
Totale	-	**1.8**	-

Cost of the OPTIMAL project

The overall cost of **OPTIMAL** is **224 k€**, of which **124 k€** are in-kind contributions, and **100 k€** are requested for the project: **72 k€** for the first year and **28 k€** for the second year of the project.

Year	Activity	Cost in Euro	In-Kind Cost in Euro
Year I	XES setup	0	120,000
Year I	Pixel Detector*	48,000	0
Year I	Motors*	10,000	0
Year I	LNF support	1,000	0
Year I	Workstation*	8,000	0
Year I	Travels	5,000	0
Total Year I	-	**72,000**	**120,000**
Year II	Portable GPU*	3,000	0
Year II	Travel to AXIm	0	4,000
Year II	Target elements	2,000	0
Year II	Publications	3,000	0
Year II	Conferences	5,000	0
Year II	Travels	5,000	0
Year II	Workshop	10,000	0
Total Year II	-	**28,000**	**4,000**
Total Cost	-	**100,000**	**124,000**



- **In-kind:**
  - XES Setup
  - Travel to the AXIm Group
- **Requested:**
  - Pixel Detector Integration
  - Computing Resources
  - Dissemination

Why Me?

- 🎓 **PhD Copenhagen:**
  - Ab initio simulations
  - ML and Databases

- 🧪 **Assegno LNF:**
  - XES measurements

nature › npj computational materials › articles › article

Article | Open access | Published: 04 March 2023

Exploring and machine learning structural instabilities in 2D materials

Simone Manti ✉, Mark Kamper Svendsen, Nikolaj R. Knøsgaard, Peder M. Lyngby & Kristian S. Thygesen

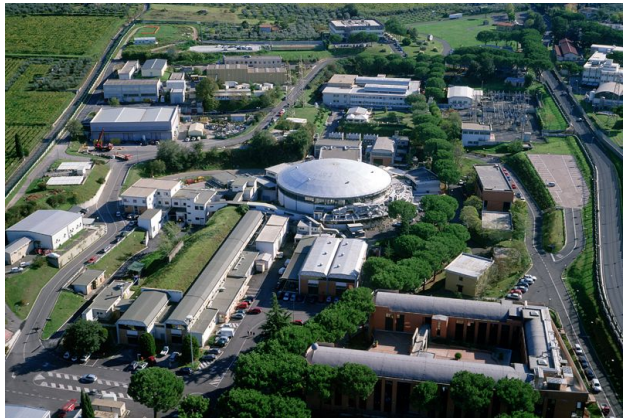
npj Computational Materials **9**, Article number: 33 (2023) | Cite this article

3469 Accesses | **15** Citations | Metrics

Here and Now

- 📍 **Current LNF** possible applications:
  VOXES, DAΦNE-Light, EuAPS and SPHINX

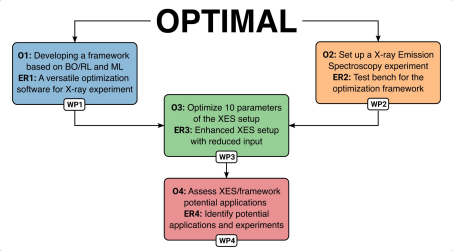
- 🚀 **Future LNF** application: EuPRAXIA



OPTIMAL for X-ray Beamline Parameters Optimization

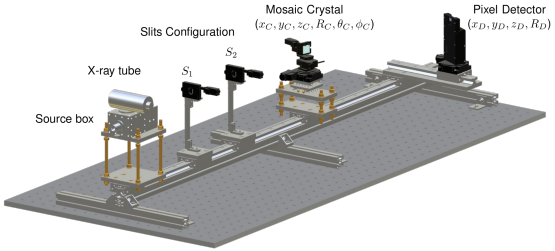
- ## 🛠️ Framework Development:

  Developing a versatile framework to optimize X-ray beamline parameters using **BO/RL** and **ML** models trained on **Ray-Tracing simulations**.



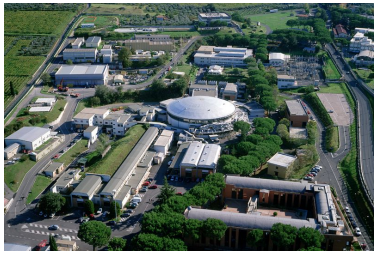
- ## 🧪 OPTIMAL Setup:

  Realize an advanced and versatile **OPTIMAL Setup** with a **faster acquisition time**, for a broad spectrum of applications inside and outside of **LNF.**



- ## 🌍 Applications:

  Propose beamline applications starting with **LNF's DAFΦNE-Light**, extending to **AXIm (London)**, and future projects like **EuPRAXIA**.



SPARE

Risks Assessment

Inaccuracy of Ray-Tracing Simulations (T1.3)

- **Risk:** Simulations may not fully represent the XES experiment due to limitations in modeling mosaic crystals.
- **Mitigation:** Collaborate with SHADOW-XOP developer Manuel Sanchez del Rio to enhance simulation accuracy.

Data Integration and Feature Extraction Issues (T1.5)

- **Risk:** Challenges in integrating data and extracting XES spectral features with deep learning.
- **Mitigation:** Leverage historical data to align simulated and real-world data.

Delays in Equipment Integration (T2.1, T2.2, T2.3)

- **Risk:** Delays in acquiring or integrating new equipment could impact timelines.
- **Mitigation:** Use a VOXES CCD as a temporary solution until the pixel detector is ready.

Optimization Complexity (T3.1, T3.2)

- **Risk:** Multi-objective optimization (e.g., time, precision, resolution) is complex and computationally demanding.
- **Mitigation:** Adjust strategies based on results and consult experts like Lucio Anderlini for AI-driven enhancements.

In-kind Cost of the OPTIMAL project

The in-kind contributions are:

- **120 k€** from VOXES project for the XES setup which comprehends:
  - **X-ray tube (10k€)**
  - **Bragg's crystal (15k€)**
  - **Bunker (5k€)**
  - **Motorized Stages (15k€)**
  - **Axis-motors (15k€)**
  - **Strip detector (25k€)**
  - **CCD (35k€)**
- **Travel to AXIm:** Travel to the AXIm group in London, for discussion with Silvia Picciccia (**4k€**)